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Studierichting: Informatica
Oefeningenreeks 4A nr. 60.14

$$\int \frac{6x+2-\sqrt{4x+1}}{2x+1+\sqrt{4x+1}}dx \overset{t=\sqrt{4x+1}}{=} \frac{1}{2}\int \frac{t\left(\dfrac{3\left(t^2-1\right)}{2}-t+2\right)}{\dfrac{t^2-1}{2}+t+1}dt$$

$$\frac{1}{2}\int \frac{3t^3-2t^2+t}{t^2+2t+1}dt = \frac{1}{2}\int \left(3t-8+\frac{14t+8}{\left(t+1\right)^2}\right)dt$$

$$= \frac{3}{4}t^2-4t+\int \frac{7t+4}{\left(t+1\right)^2}dt = \frac{3}{4}t^2-4t+7\int \frac{t+1-1}{\left(t+1\right)^2}dt+4\int \frac{1}{\left(t-1\right)^2}dt$$

$$= \frac{3}{4}t^2-4t+\frac{3}{t+1}+7\ln|t+1|+c$$

$$\overset{t=\sqrt{4x+1}}{=} 3x-4\sqrt{4x+1}+\frac{3}{\sqrt{4x+1}+1}+7\ln\left|\sqrt{4x+1}+1\right|+c$$

References